

# Solution Architecture

ComicCraft - AI Comic Story Creator using Gemini Models  
Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester  
Project period: 1 Oct 2026 - 2 Oct 2026

Browser: Jinja2 pages

FastAPI app (main.py, routes.py)

gemini\_flash.py



gemini\_pro.py



image\_generator.py

layout\_builder.py



exporters.py

static/panels | static/exports

## Components

| Component              | Responsibility                                           |
|------------------------|----------------------------------------------------------|
| app/main.py            | Creates app, mounts /static, starts model warm-up thread |
| app/routes.py          | Routes and pipeline orchestration, error handling        |
| app/gemini_flash.py    | Outline generation in JSON mode                          |
| app/gemini_pro.py      | Story text in fixed Panel/Caption/Narration format       |
| app/image_generator.py | Cached Stable Diffusion pipeline; saves PNG; returns URL |
| app/layout_builder.py  | Regex-parses story; merges with images                   |
| app/exporters.py       | FPDF export, one page per panel                          |
| templates/             | index, comic_preview, export_success                     |
| static/                | bg.jpg, fonts, panels, exports                           |

## Routes

| Route                | Method | Purpose                    |
|----------------------|--------|----------------------------|
| /                    | GET    | Home form                  |
| /generate            | POST   | Run pipeline, show preview |
| /generate-comic/json | POST   | JSON result                |
| /export-success      | GET    | Download confirmation      |

|             |     |                   |
|-------------|-----|-------------------|
| /test-image | GET | Single test image |
|-------------|-----|-------------------|

**External dependencies**

- Google Gemini API over HTTPS (needs GEMINI\_API\_KEY)
- Hugging Face hub for first-time model download, then local cache